and nearest neighbor search to find the closest hub for deliveries, so that parcels need to travel the least possible distance. Nevertheless, there will still be some road to hit to get our orders to the customers, and our margin gets thinner if we waste time and gas for each delivery.

Conversely, if we were able to optimize the route fared, we could save some money for each parcel, and a lot of money at scale, on thousands or millions of deliveries.

In this chapter we will solve this problem, finding the best route to make a single delivery (from a warehouse to the customer's home), by using graphs; we will tackle the problem at different levels of abstraction, demonstrating how search algorithms like BFS, Dijkstra and A* work.

14.1 Definitions

A graph G is usually defined in terms of two sets:

- A set of *vertices* `V`: independent, distinct entities that can appear in any multiplicity. A graph can have 1, 2, 100 or any number of vertices but, in general, graphs don't support duplicate vertices.
- A set of *edges* `E` connecting vertices: an edge is a pair of vertices, the first one usually denoted as the *source* vertex, and the second one called the *destination* vertex.

So, we write `G=(V,E)` to make it clear that the graph is made of certain sets of vertices and edges; for instance, the graph in figure 14.2 is formally written as